Bhargav Hegde

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AI Engineer specializing in Retrieval-Augmented Generation (RAG), V2X Cooperative Perception, and Computer Vision with 3+ years of production software engineering experience

Education

State University of New York at Buffalo, Buffalo, NY Aug. 2024 – Present
Master of Science in Computer Science and Engineering
Research Associate in CAVAS Lab (Connected and Autonomous Vehicle Applications and Systems) - conducting research on real-time 3D object detection, LiDAR-based perception, and V2X cooperative perception for autonomous vehicles

JSS Academy of Technical Education, Bangalore, India Aug. 2016 – Aug. 2020
Bachelor of Engineering in Computer Science and Engineering
Initiated and **secured funding** for an IoT laboratory; conducted instructional sessions for junior students

Professional Experience

Tech Mahindra, Software Engineer Nov. 2021 – Jul. 2024
Developed automation frameworks using Robot Framework, applying machine learning-based pattern recognition for Wi-Fi RTT locationing systems (IEEE 802.11mc), achieving sub-meter accuracy through advanced packet analysis with Wireshark.
Led R&D initiatives in DHCPv6 protocol testing, designed comprehensive test cases addressing stateful vs. stateless configurations, and delivered training to cross-functional teams globally on Wi-Fi RTT implementation.

Tata Consultancy Services, System Associate Engineer Apr. 2021 – Oct. 2021
Developed Python-based RESTful APIs to automate business processes, streamlining data fetching, processing, and storage for efficient client reporting, accelerating deployment time by 15%.

Internships 2018 – 2020
Pentagon Space (2020): Completed 4-month Python Full Stack course; built web applications with Django and SQL.
Teqed Labs (2019): Developed Computer Vision-based attendance system using Deep Learning for face recognition and database management — **Awarded 'Best Project'**.
Experts Hub (2018): Built machine learning pipeline to classify plant and seedling images with 92% accuracy on 10,000+ image dataset. Led and mentored team of 8 members — **Awarded 'Best Intern'**.

Projects

DAIR-V2X Cooperative LIDAR 3D Object Detection (PyTorch, CUDA, V2X) 2025
Full reproduction of DAIR-V2X late-fusion model for cooperative 3D object detection in V2X scenarios, achieving 40.01% AP on vehicle class with improved performance metrics: 37.25% 3D AP@0.7 (8.9% gain).
Reduced communication cost by 12.3% to 898.10 bytes; implemented LiDAR point cloud processing with visualization tools for 3D and BEV fusion results.

KnowBot 3.0: Neural Knowledge Assistant (Next.js 15, Django DRF, LangChain, Hybrid Search) 2025
Architected a high-fidelity, decoupled RAG system featuring an advanced **Hybrid Search** engine (BM25 + ChromaDB) and intelligent context window management for superior retrieval accuracy.
Developed an autonomous **Multi-modal Pipeline** integrating **Tesseract OCR** for scanned document ingestion and **Tavily Web Search** for real-time knowledge expansion and query reformulation.
Implemented as a cloud-native platform with **microservices architecture** (Celery, Redis, PostgreSQL), featuring a premium 3D-styled UI and custom system personas.

KnowBot 2.0: Personal RAG Chatbot (Python, Streamlit, Ollama, LangChain, ChromaDB) 2024
Developed a privacy-first, fully local RAG application for interacting with personal documents (PDF, TXT, MD) using Llama 3.1 and nomic-embed-text via Ollama.
Integrated source citations, persistent vector storage, and hallucination guardrails to ensure data security and response reliability in 100% offline environments.

Garbage Segregation System (YOLOv3, CNN, Raspberry Pi, Robotic Arm) Jun. 2020
Designed robotic waste segregation system using YOLOv3 and custom CNNs, achieving 90% garbage classification accuracy with optimized edge inference on Raspberry Pi.
Automated waste sorting and placement with integrated robotic arm; **secured funding from Karnataka State Council (KSCST)**.
For a complete list of projects, please visit: bhargavhegde.github.io

Technical Skills

Technical Skills

Programming Languages: Python, C/C++, Java, SQL, JavaScript/TypeScript, Next.js, HTML/CSS
AI/ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, YOLO, LangChain, Ollama, Tesseract
Deep Learning: Computer Vision, NLP, Reinforcement Learning, LLMs (Llama 3.1), RAG, Hybrid Search, OCR
Specialized Tech: LiDAR Processing, 3D Object Detection, V2X, CUDA, Embedded Systems (Raspberry Pi), Tavily API
Tools & Platforms: Docker, Git, Wireshark, Robot Framework, Vercel, Railway, ChromaDB, Redis, Celery, AWS
Databases & Networking: MySQL, PostgreSQL, Linux/UNIX, IoT, IEEE 802.11mc, DHCPv6, IPv6, Wi-Fi RTT